THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS · HARD LEVEL CHAPTER TEST SET 3

Chapter 1: **A Square and a Cube** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The number of digits in the square root of a 7-digit perfect square is: 1

(a) 4 **(b)** 3 **(c)** 6 **(d)** 5

2. $\sqrt[3]{-2197}$ is: 1

(a) 13 **(b)** -14 **(c)** -13 **(d)** 14

3. The ones digit of 349^3 is: 1

(a) 7 **(b)** 1 **(c)** 3 **(d)** 9

4. The value of $\sqrt{961} + \sqrt{256}$ is: 1

(a) 496 **(b)** 15 **(c)** 48 **(d)** 47

5. The number of perfect squares between 150 and 1250 (both excluded) is: 1

(a) 23 **(b)** 22 **(c)** 25 **(d)** 24

6. The smallest number to be added to 4567 to make it a perfect square is: 1

(a) 57 **(b)** 56 **(c)** 59 **(d)** 58

7. The largest number to be subtracted from 9018 to make it a perfect square is: 1

(a) 185 **(b)** 183 **(c)** 182 **(d)** 184

8. The difference of the squares of two consecutive natural numbers is 149. The numbers are: 1

(a) 74 and 76 **(b)** 73 and 74 **(c)** 75 and 76 **(d)** 74 and 75

9. **Assertion (A):** The number of non-square numbers between 20^2 and 21^2 is 40. 1
Reason (R): There are $2n$ non-square numbers between n^2 and $(n+1)^2$.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. **Assertion (A):** The cube root of -27 is -3. 1
Reason (R): A negative number has no cube root.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Find the number of non-square numbers between 11^2 and 14^2. 2

12. Find the smallest number to be added to 9800 to make it a perfect square. What is the square root of the result? 2

13. The difference between the squares of two consecutive numbers is 171. Find the numbers. 2

14. Find the number of non-square numbers between 13^2 and 16^2. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A square garden has area 5625 m². Fencing costs ₹20 per metre. Find the total cost of fencing it. 3

16. A solid cube of side 11 cm is cut into small cubes of side 1 cm. How many small cubes are formed? How many have exactly three painted faces if the big cube was painted all over? 3

17. A rectangular hall 60 m by 84 m is to be paved with equal square tiles, the largest possible. Find the side of each tile and the number of tiles. 3

OR

A square garden has area 7056 m². Fencing costs ₹15 per metre. Find the total cost of fencing it. 3

18. A solid cube of side 12 cm is cut into small cubes of side 1 cm. How many small cubes are formed? How many have exactly three painted faces if the big cube was painted all over? 3

OR

A rectangular hall 66 m by 88 m is to be paved with equal square tiles, the largest possible. Find the side of each tile and the number of tiles. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Case study. Meera notices that $1 + 3 + 5 + 7 + 9 + 11 + 13 = 49$. She wonders if odd numbers can help her decide whether a number is a perfect square. Her teacher also gives her the number 2000. 5

(i) How many odd numbers were added to get 49?

(ii) Find the sum of the first 15 odd natural numbers.

(iii) Is 2000 a perfect square? Give a reason.

(iv) Find the smallest number by which 2000 must be divided to get a perfect square, and the square root of the quotient.

20. Find the smallest square number that is divisible by each of 15, 18 and 24. 5

OR

Find the smallest cube number that is divisible by each of 12, 20 and 45. 5

END OF QUESTION PAPER — Total: 40 Marks